

Section 1.1: Slopes and Equations of Lines

Finite Mathematics MTH 132 Spring 2014

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Johnson C. Smith University

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Chapter 1 Outline

Chapter 1: Linear Functions

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1.1 Slopes and Equations of Lines

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1.2 Linear Functions & Applications

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1.1 Slopes and Equations of Lines

1.2 Linear Functions & Applications

1.3 Least Squares Line

Linear Equations/Functions

Linear Equations/Functions

What is a linear functions?

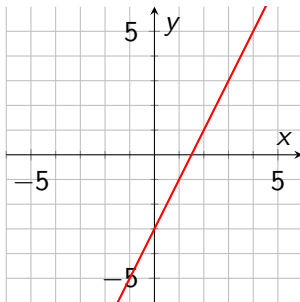
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Linear Equations/Functions

What is a linear functions?

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ex. $y = 2x - 3$



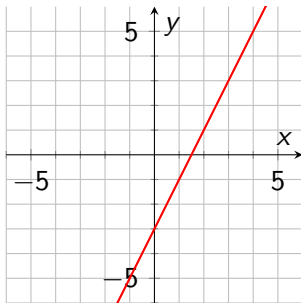
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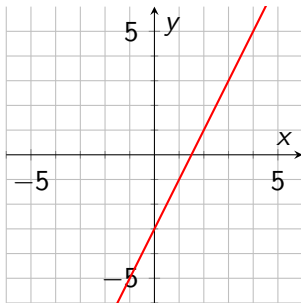
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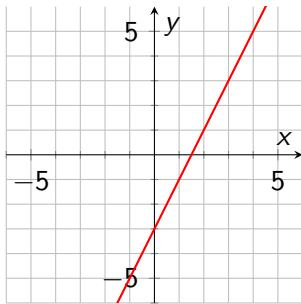
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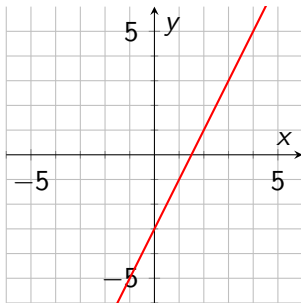
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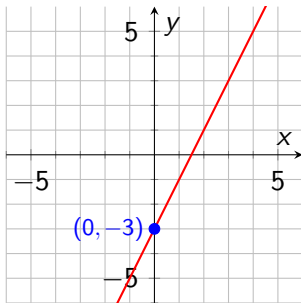
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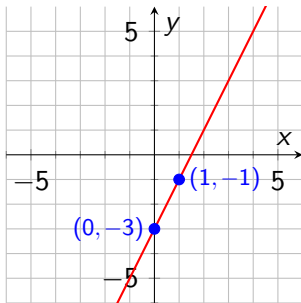
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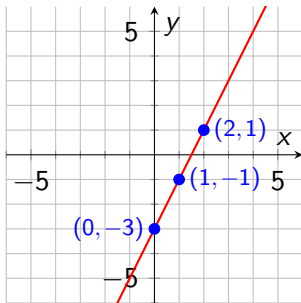
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Point-Slope Formula

$$y - y_1 = m(x - x_1)$$

Find the equation of a Line

Slope-intercept Form

$$y = mx + b$$

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Example 1.1.4.

Find the slope-int. form of the line with a slope of -3 passing through $(6,6)$.

Find the equation of a Line

Slope-intercept Form

$$y = mx + b$$

Point-Slope Formula

$$y - y_1 = m(x - x_1)$$

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Example 1.1.5.

find the equation of the line which passes through $(-2, 5)$ and $(3, 6)$.

Find the equation of a Line

Slope-intercept Form

$$y = mx + b$$

Point-Slope Formula

$$y - y_1 = m(x - x_1)$$

- m is the slope.
- b is the y -int.

Example 1.1.6.

Find the equation of the line that is *parallel* to $y = x + 1$ and passes through $(0, 2)$.

Find the equation of a Line

Slope-intercept Form

$$y = mx + b$$

Point-Slope Formula

$$y - y_1 = m(x - x_1)$$

- m is the slope.
- b is the y -int.

Example 1.1.7.

Find the equation of the line that is *perpendicular* to $y = \frac{2}{3}x + 1$ and passes through $(0, 2)$.

Summary

Equation	Description
$y = mx + b$	Slope-int. form slope m , y-int b
$y - y_1 = m(x - x_1)$	Pt-slope formula passing through (x_1, y_1)
$y = k$	Horizontal line through (x, k)
$x = k$	Vertical line through (k, y)
$m = \frac{a}{b}$	if parallel to $y = \frac{a}{b}x + b$
$m = -\frac{b}{a}$	if perpendicular to $y = \frac{a}{b}x + b$

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Ex. 1.1.1. Use the conditions to write an equation for the line in slope-int. form.

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Ex. 1.1.1. Use the conditions to write an equation for the line in slope-int. form.

Slope = -3 , m passing through $(6, 6)$.

Ex. 1.1.2. Use the conditions to write an equation for the line in slope-int. form.

Perpendicular to $y = -\frac{1}{2}x + 7$ passing through $(7, 3)$.

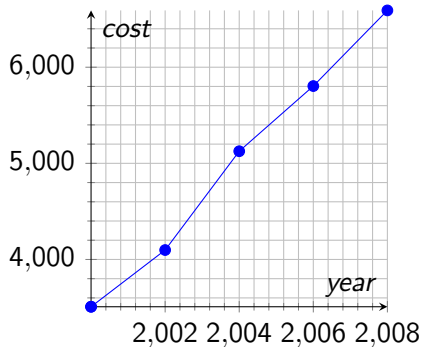
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